

Yixuan Wu

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Aug. 2024 to Dec. 2025 Expected

Master of Engineering in Financial Engineering, **GPA: 4.3/4.3**

Relevant Coursework: Machine Learning, Financial Engineering with Stochastic Calculus, Market Microstructure and Trading, Optimization Modeling in Finance, Quantitative Method of Risk Management, Monte Carlo Simulation, Big Data Technologies

University of Waterloo, Waterloo, ON, CA

Sep. 2020 to May 2024

Bachelor of Mathematics in Mathematical Finance & Statistics, Minor in Pure Mathematics, **GPA: 93.44/100**

Relevant Coursework: Time Series, Generalized Linear Model, Computational Statistics & Data Analytics, Measure Theory

Honors: Third-highest GPA in Major, Faculty of Mathematics Senate Scholarships, Pure Math Department Silver Award

PROFESSIONAL EXPERIENCE

Quantitative Research Intern, *First State Cinda Fund Management*, Shanghai, China

Dec. 2024 to Jan. 2025

- Developed a multi-objective genetic programming framework for factor mining, applying IC and NDCG fitness metrics and leveraging NSGA-II to select Pareto-optimal factors while preserving population diversity
- Designed 64 genetic operators, including cross-sectional and time-series operators, to boost factor explanatory power, and validated their long-term robustness via out-of-sample screening
- Synthesized a composite factor from the top 10 IR-ranked factors, yielding 21.7% annualized excess return, 1.45 Sharpe ratio, and -23.1% max drawdown across validation and test sets

Data Analytics Intern, *Guosheng Security*, Shanghai, China

July 2023 to Aug. 2023

- Built classification models to extract predictive price-movement signals from financial news, using logistic regression, random forest, XGBoost, and LSTM, achieving over 72% accuracy in forecasting short-term stock price direction
- Executed time series similarity analysis on thousands of fund NAV curves using DTW and various similarity metrics, ranking funds by resemblance to a target one, and creating a pool for alternative fund selection and risk diversification
- Engineered an ETL pipeline to integrate client trade data with historical trends, generating trading preference labels to support personalized strategy recommendations for wealth management clients

Risk Management Intern, *Fullgoal Fund Management*, Shanghai, China

May 2023 to July 2023

- Automated daily risk monitoring workflows in Python by analyzing a dataset of 4,000+ stocks, generating real-time compliance alerts and market risk insights for fund managers
- Improved reconciliation accuracy by refining variance analysis algorithms, identifying mismatches between financial reports and actuals for semi-annual audits

PROJECTS

Limit Order Execution Algorithm Design (team of 4), *Cornell University*, Ithaca, NY

Mar. 2025 to Present

- Conducted exploratory analysis on high-frequency limit order book data across five equities; engineered and validated predictive microstructure features through various visualization and event studies
- Formulated an execution strategy by extracting favorable trading instances from historical limit order book data to train a machine learning classifier, and applied the classifier to real-time market features to decide whether to execute trades
- Built a real-time trade monitoring system with a fallback mechanism to ensure order completion, and evaluated performance against a TWAP baseline to assess execution quality

LVaR Portfolio Optimization with Dynamic Liquidity Constraints, *Cornell University*, Ithaca, NY

Feb. 2025 to Present

- Constructed a portfolio optimization framework applying liquidity-adjusted Value-at-Risk (LVaR) to capture execution risk under extreme market conditions better
- Incorporated regime-aware, sector-specific liquidity constraints by calibrating liquidation time using volume data and market volatility (VIX), enabling dynamic reallocation under shifting environments
- Backtested the strategy across COVID and post-COVID periods, benchmarking against traditional mean-variance portfolios to validate performance under real-world liquidity shocks

SKILLS, ACTIVITIES & INTERESTS

- Technical Skills:** Python (Pandas, NumPy, sklearn, Statsmodels, deap), R, Matlab, Excel, LaTeX
- Interests:** Debating, Poker, Nature Photography, Musicals